



Armand Bester

Passionate about interdisciplinary science

- ▶ Bloemfontein, South Africa
- ▶ South African
- ▶ Married

Skills - data

Python 10+ yrs



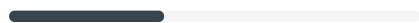
Bash 10+ yrs



R 5 yrs



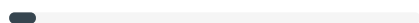
C++ (Arduino) 5 yrs



Rmarkdown 2 yrs



Quarto 4 yrs



Analysis of NGS data 10+ yrs



Data Analysis 10+ yrs



Internet of Things 2 yrs



Skills - lab

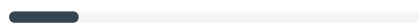
Sanger sequencing 10+ yrs



Oxford NanoPore sequencing 6 yrs



IonTorrent sequencing newbie



Summary

The rapid growth of scientific data throughout my career highlighted the increasing importance of computational approaches in modern research. I have always been fascinated by algorithms and programming, and the intersection of informatics and science is where my interests lie. I enjoy developing computational solutions to extract meaningful insights from complex biological data, and I value the opportunity to contribute to a diverse range of interdisciplinary research projects.

Experience

Senior Medical Scientist

National Health Laboratory Services

Aug 2021 - Present

I am a practising medical scientist in the Division of Virology where I am involved with both diagnostics and research.

Medical Scientist

National Health Laboratory Services

Jan 2015 - Jul 2021

Assistant Researcher

Dept. Medical Virology

Jan 2011 - Dec 2014

During this time I developed an HIV drug resistance assay which was later implemented

Research Assistant

National Earthquake Laboratory of South African Mines

Jan 2007 - Dec 2010

Running and maintaining a quadrupole mass spectrometer for measuring gasses from rock 3.6 km deep

Projects

PhyloPi

An affordable, purpose-built phylogenetic pipeline for the HIV drug resistance testing facility

HIV drug resistance testing involves sequencing some parts of the viral genome, which informs us of the virus's susceptibility to Anti-Retroviral drugs. If the patient shows resistance to a particular drug they are taking, this guides the clinician in adjusting the regimen. Additionally, when generating this sequence data one can do phylogenetic analysis as quality control. This led us to develop a standalone bioinformatic pipeline. Apart from the academic adventure this also gave us the opportunity to write about HIV on RStudio's blog post (<https://rviews.rstudio.com/2019/04/30/analysing-hiv-pandemic-part-1/>). This four-part series shows the analysis of public data using R and was a fun outreach and I am quite proud about the people I got to work with.

Molecular recombinant technology 10+ yrs

PCR and real-time PCR design 10+ yrs

Site directed mutagenesis and cloning 10+ yrs

Tools

- ▶ Jupyter
- ▶ Terminal
- ▶ Git
- ▶ RStudio
- ▶ VS Code

Professional registration

1 Dec 2014

MS 0001511

HPCSA

Education

2007

M.Sc Biochemistry

University of the Free State

2004

B.Sc Honours in Biochemistry

University of the Free State

2003

Sequencing, ecology and surveillance

Data generation and analysis. Sequencing, bash, python

The combination of next-generation sequencing technologies and bioinformatics provides opportunities to collaborate on a diverse range of research projects. These include viral surveillance, characterising the virome and microbiome of the infant gut, investigating fungal communities in urban water systems, and applying sequencing-based approaches to a wide variety of biological and public health research questions.

Hepatitis, HIV and other co-infections

Python, data wrangling, infographics, statistics, R

Large-scale patient laboratory data are analysed using Python and R to investigate the epidemiology of Hepatitis B virus (HBV) infection, including HIV and Hepatitis C virus (HCV) co-infections. The work aims to better understand the burden of this highly infectious yet vaccine-preventable disease. In parallel, a systematic review and meta-analysis of HBV serology across Africa has been undertaken, accompanied by the development of an interactive dashboard to visualise and explore the findings. (<https://hbv-geo.shinyapps.io/oxafricahbv/>).

A microgrid in an energy crisis

Influxdb, telegraf, grafana

Working with the brilliant people at the GRRG (Grid Related Research Group, UFS) and the eResearch High Performance Computing unit, we developed a dashboard showing real-time power consumption and grid frequency (<https://s.ern.ufs.ac.za/grrg>) for both the Bloemfontein as well as QwaQwa campuses. We are using data to manage resource planning for power demand and, together with the ICDF (Interdisciplinary Centre for Digital Futures), are trying to understand how social layers and the usage and understanding of electricity influence each other.

Pathogen Research Laboratory

Screening, surveillance, metagenomics

- Screening of vectors for arboviruses in South Africa.
- Screening cattle and sheep sera for evidence of infection with arboviruses identified in Bloemfontein.
- Next Generation Sequencing (NGS) for complete genome determination of arboviruses and zoonotic pathogens.
- Development of Oxford NanoPore sequencing techniques.
- Bayesian phylogeography of global Crimean-Congo hemorrhagic fever virus isolates.
- Arboviruses as a cause of neurological disease in humans.
- Detection and whole genome characterization of Jingmen tick virus from wild-caught ticks (Acari: Ixodidae) in the Free State and Eastern Cape Province South Africa.

Teaching experience at NHLS/UFS

- Main and co- supervision for Honours and M. students and collaborations.
 - Co-supervisor for M.Med.Sc student. Detection and characterization of alphaviruses in the Bloemfontein region, Free State.
 - Co-supervisor for M.Med.Sc student. Development of Non-Species-Specific ELISA and RT-qPCR for Serosurveillance and Detection of Crimean-Congo Haemorrhagic Fever Virus.

B.Sc Biochemistry and Organic Chemistry

University of the Free State

Contact

📍 Bloemfontein, South Africa

☎ +27 72 293 7193

✉ besterpa.sci@gmail.com

👤 Armand Bester

🆔 Armand Bester

- Collaborator on Ph.D project. Circulation and characterization of known and novel orthonairoviruses transmitted by ixodid ticks in three provinces of South Africa.
- Collaborator on Ph.D project. Genomic characterization of *Mycobacterium* spp. Isolated from cattle using long-read Oxford nanopore sequencing.
- Collaborator on Ph.D project. Determination of tick diversity and screening for viral infections in ticks of livestock from the Eastern Cape and Free State Provinces of South Africa.
- Co-supervisor (Norwegian University of life sciences) M.Sc student. Forecasting load in an operational centralised microgrid.
- Main supervisor for Honours student developing an HIV integrase drug resistance assay.
- Equally shared supervisor for two Honours students on: A simple and rapid approach to prepare Sindbis and West Nile viral RNA controls for differentiation between positive samples and laboratory contamination.
- Co-supervisor for M. Med student looking at the influence of volumes on dry blood spot samples for infant HIV diagnostics.
- Co-supervision of M. Med student looking at HBV drug resistance and viral loads.
- Collaborator on M. Med project. Hepatitis B seroprevalence amongst HIV infected individuals in the Free State, South Africa
- Collaborator on M. Med project. HBV viral load and drug resistance among HIV-HBV co-infected patients: A cross-sectional study in central South Africa.
- Co-supervision of Honours student in the development of RT-qPCR to detect and differentiate between orthobunyaviruses.
- Co-supervision of Honours student in the development of a VWF domains A1, A2 and A3 single molecule sequencing assay.
- Screening black South African females with Type 2 Diabetes Mellitus for mutations in the peroxisome proliferator-activated receptor gamma gene.
- Methylation analysis of type two diabetes mellitus (T2DM) genes in cases and controls from central South Africa.
- SOCS3 gene DNA methylation analysis methods to investigate inflammation associated with HIV infection.
- Marking of Honours and M. students dissertations.
 - Marking of M. Sc thesis for Prof Gert van Zyl, Stellenbosch University
- Lecturing
 - Honours students and medical registrars in bioinformatics, an introduction (<https://armandbester.github.io/bioinf/>).
 - Honours students and medical registrars, qPCR theory and practical.

Peer reviewed publications

- Arbovirus surveillance detects mosquito-borne viruses in central South Africa. **Virology**. 2025. <https://doi.org/10.1016/j.virol.2025.110668>
- Identification of arboviruses in mosquito populations in KwaZulu-Natal, South Africa and the first record of *Wyeomyia mitchellii* in the Old World. **PLOS Neglected Tropical Diseases**. 2025. 19(8): e0013093. <https://doi.org/10.1371/journal.pntd.0013093>

- Hepatitis B seroprevalence among HIV-positive adults in the Free State, South Africa. **S Afr J HIV Med.** 2025; 26(1), a1654. <https://doi.org/10.4102/sajhivmed.v26i1.1654>
- Culture-Dependent and -Independent Wastewater Surveillance for Multiple Pathogenic Yeasts. **Journal of Fungi.** 2025; 11(2):86. <https://doi.org/10.3390/jof11020086>
- Biofilms on urban aquatic plastic pollution as a reservoir for pathogenic yeasts. **Journal of Water and Health.** 2024 Oct <https://doi.org/10.2166/wh.2024.133>.
- Whole genome molecular analysis of respiratory syncytial virus pre and during the Covid-19 pandemic in free state province, South Africa. **Virus Res.** 2024 Jun 26:199421. doi: 10.1016/j.virusres.2024.199421. Epub ahead of print. PMID: 38942296.
- Longitudinal analysis of the enteric virome in paediatric subjects from the Free State Province, South Africa, reveals early gut colonisation and temporal dynamics. **Virus Res.** 2024 May 31;346:199403. doi: 10.1016/j.virusres.2024.199403.
- Multiplex PCR method for MinION sequencing of Bagaza virus isolated from wild caught mosquitoes in South Africa. **Journal of Virological Methods.** Volume 327, 2024, 114917, ISSN 0166-0934, <https://doi.org/10.1016/j.jviromet.2024.114917>
- Data-Driven Modeling of Frequency Dynamics Observed in Operating Microgrids: A South African University Campus Case Study. 2024. **IEEE Access.** <http://dx.doi.org/10.1109/ACCESS.2024.3357945>
- Advancing HIV Drug Resistance Technologies and Strategies: Insights from South Africa's Experience and Future Directions for Resource-Limited Settings. 2023. **Diagnostics.** <https://doi.org/10.3390/diagnostics13132209>
- Sindbis Virus Antibody Seroprevalence in Central Plateau Populations, South Africa. 2022. **Emerging Infectious Diseases.** <https://doi.org/10.3201%2Feid2810.211798>
- SARS-CoV-2 variants from COVID-19 positive cases in the Free State province, South Africa from July 2020 to December 2021. 2022. **Frontiers in Virology** <https://doi.org/10.3389/fviro.2022.935131>
- A decontamination strategy for resolving SARSCoV2 amplicon contamination in a next generation sequencing laboratory. 2022. **Archives of Virology.** <https://doi.org/10.1007%2Fs00705-022-05411-z>
- Emergence of SARS-CoV-2 Omicron lineages BA.4 and BA.5 in South Africa. 2022. **Nature Medicine.** <https://doi.org/10.1038/s41591-022-01911-2>
- Rapid epidemic expansion of the SARS-CoV-2 Omicron variant in southern Africa. 2022. **Nature.** <https://doi.org/10.1038/s41586-022-04411-y>
- Whole-Genome Sequence and Comparative Analysis of Human Papillomavirus Type 18 Isolated from a Nasopharyngeal Carcinoma from South Africa. 2021. **Microbiology Resource Announcements.** <https://doi.org/10.1128/MRA.00630-21>
- Complete genome sequence of a HPV31 isolate from laryngeal squamous cell carcinoma and biological consequences for p97 promoter activity. 2021. **PLoS ONE.** <https://doi.org/10.1371/journal.pone.0252524>
- Rare BRCA2 Variants With Founder Haplotypes in the South African Population: Implications for Point-of-Care Testing Based on a Single-Institution BRCA1/2 Next-Generation Sequencing Study. 2021. **Frontiers in Oncology.** <https://doi.org/10.3389/fonc.2020.619469>
- Climatic and topographic changes since the Miocene influenced the diversification and biogeography of the tent tortoise (*Psammobates tentorius*) species complex in Southern Africa. 2020. **BMC Evolutionary Biology.** <https://doi.org/10.1186/s12862-020-01717-1>

- Metagenomic Analysis of the Enteric RNA Virome of Infants from the Oukase Clinic, North West Province, South Africa, Reveals Diverse Eukaryotic Viruses. 2020. **Viruses** <https://doi.org/10.3390/v12111260>
- Characteristics of individuals with hepatitis B virus infection in the Free State and Northern Cape provinces of South Africa. 2020. **International Journal of Infectious Diseases**. DOI:<https://doi.org/10.1016/j.ijid.2020.09.1363>
- Persistence of Crimean- Congo Hemorrhagic Fever Virus RNA. 2020. **Emerging Infectious Diseases**. <https://dx.doi.org/10.3201/eid2602.191460>.
- A simple and rapid approach to prepare Sindbis and West Nile viral RNA controls for differentiation between positive samples and laboratory contamination. 2020. **Journal of Virological Methods**. <https://doi.org/10.1016/j.jviromet.2020.113822>.
- Unraveling the diversification and systematic puzzle of the highly polymorphic *Psammobates tentorius* (Bell, 1828) complex (Reptilia: Testudinidae) through phylogenetic analyses and species delimitation approaches. 2020. **Journal of Zoological Systematics and Evolutionary Research**. <https://doi.org/10.1111/jzs.12338>
- Hepatitis B virus seroepidemiology data for Africa: Modelling intervention strategies based on a systematic review and meta-analysis. 2020. **PLoS Med**. <https://doi.org/10.1371/journal.pmed.1003068>
- PhyloPi: An affordable, purpose built phylogenetic pipeline for the HIV drug resistance testing facility. 2019. **PloS ONE**. <https://doi.org/10.1371/journal.pone.0213241>
- Seroepidemiologic Survey of Crimean-Congo Hemorrhagic Fever Virus in Selected Risk Groups, South Africa. 2018. **Emerging Infectious Diseases**. <https://doi.org/10.3201%2Feid2407.172096>
- Determination of the complete genome and functional analysis of HPV6 isolate VBD19/10 from a patient with aggressive recurrent respiratory papillomatosis. 2016. **Epidemiology and Infection**. <https://doi.org/10.1017/s0950268816000388>
- Global epidemiology of drug resistance following failure of WHO recommended first line regimens for adult HIV-1 infection – a multi-centre retrospective cohort study. 2016. **The Lancet Infectious Diseases**. [https://doi.org/10.1016/s1473-3099\(15\)00536-8](https://doi.org/10.1016/s1473-3099(15)00536-8)
- Comparative analysis of the L, M, and S RNA segments of Crimean-Congo haemorrhagic fever virus isolates from southern Africa. 2015 **Journal of medical virology**. <https://doi.org/10.1002/jmv.24079>
- Next-generation sequencing of southern African Crimean-Congo haemorrhagic fever virus isolates reveals a high frequency of M segment reassortment. 2014. **Epidemiology and Infection**. <https://doi.org/10.1017/s0950268814000818>
- Southern African Treatment Resistance Network (SATuRN) RegaDB HIV drug resistance and clinical management database: supporting patient management, surveillance and research in southern Africa. 2014 **Database (Oxford)**. <https://doi.org/10.1093/database/bat082>
- Direct observation of geogas transport through an inactive fault system at 3.54 km depth, TauTona gold mine, South Africa. 2011 **Applied Geochemistry**. <https://doi.org/10.1016/j.apgeochem.2011.07.011>
- Worms from hell: Nematoda from the terrestrial deep subsurface of South Africa. 2011 **Nature**. <https://doi.org/10.1038/nature09974>
- A thioredoxin reductase-like protein from the thermophile, *Thermus scotoductus* SA-01, displaying iron reductase activity. 2010 **FEMS Microbiology Letters**. <https://doi.org/10.1111/j.1574-6968.2009.01852.x>
- Metabolic promiscuity from the deep subsurface: a story of Survival or Superiority. **Instruments, Methods, and Missions for Astrobiology XI**. Edited by Hoover, Richard B.; Levin, Gilbert V.; Rozanov, Alexei Y.; Davies, Paul C. **Proceedings of the SPIE**. <https://doi.org/10.1117/12.801142>